

ST BRANDON ILE RAPHAEL, Mauritius

WMO: 619860

Lat: 16.433S

Lon: 59.605E

Elev: 4

StdP: 101.28

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.8	22.2	16.3	11.6	23.3	17.1	12.2	23.5	12.7	23.3	12.3	23.5	6.7	120	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	4.1	32.7	27.1	32.1	26.9	31.6	26.7	27.8	31.4	27.5	31.0	27.2	30.5	3.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	22.4	30.1	26.5	22.0	29.8	26.2	21.6	29.5	88.9	31.7	87.3	31.2	85.8	30.8	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.9	11.8	10.7	DB	20.7	33.9	0.6	0.6	20.2	34.4	19.8	34.7	19.5	35.0	19.0	35.4	
			WB	17.4	28.7	1.0	0.3	16.6	28.9	16.0	29.1	15.4	29.3	14.7	29.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.8	28.7	28.8	28.9	28.2	27.0	25.4	24.7	24.4	24.7	25.6	26.9	28.1
	DBStd	1.98	1.12	1.17	0.99	0.96	1.07	1.04	0.94	0.97	0.93	1.11	1.23	1.08
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6124	581	526	586	547	527	462	455	446	442	485	506	562
	CDD18.3	3082	323	292	327	297	269	212	197	187	192	227	256	303
	CDH23.3	27478	3752	3451	3841	3238	2519	1338	873	696	820	1445	2241	3264
CDH26.7	7163	1337	1258	1399	937	446	90	34	27	46	165	457	969	
Wind	WSAvg	6.0	4.8	5.0	5.4	5.8	5.9	7.0	7.0	6.7	6.6	6.3	6.0	5.5
Precipitation	PrecAvg	954	160	187	135	102	72	51	49	47	28	28	31	68
	PrecMax	1541	424	638	409	351	305	130	131	201	67	77	118	341
	PrecMin	495	14	14	38	17	6	9	13	12	5	7	4	1
	PrecStd	232	102	139	76	72	58	28	23	30	15	15	26	62
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.6	33.4	33.2	32.2	31.3	29.3	28.5	28.3	28.7	29.8	31.6	33.0
		MCWB	27.0	27.6	27.7	27.1	26.4	24.9	23.9	23.5	23.5	24.6	25.9	27.1
	2%	DB	32.7	32.5	32.3	31.5	30.4	28.4	27.6	27.4	27.8	29.0	30.6	32.0
		MCWB	26.9	27.2	27.3	26.7	25.9	24.2	23.3	22.9	23.1	24.1	25.5	26.5
	5%	DB	31.9	31.8	31.6	30.9	29.7	27.7	26.9	26.7	27.2	28.3	29.9	31.2
		MCWB	26.7	27.0	27.0	26.4	25.5	23.8	23.0	22.5	22.8	23.7	25.2	26.1
	10%	DB	31.1	31.0	31.0	30.2	28.9	27.1	26.2	26.1	26.5	27.6	29.1	30.4
		MCWB	26.4	26.8	26.7	26.2	25.0	23.5	22.6	22.1	22.4	23.4	24.7	25.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.1	28.2	28.3	27.8	27.1	25.9	24.7	24.5	24.4	25.6	26.8	27.6
		MCDB	32.1	32.0	32.0	31.2	30.3	28.5	27.3	27.4	27.2	28.5	30.3	31.8
	2%	WB	27.5	27.7	27.8	27.3	26.5	24.9	24.1	23.6	23.7	24.9	26.1	27.1
		MCDB	31.3	31.2	31.3	30.6	29.5	27.3	26.7	26.5	26.8	27.8	29.4	31.0
	5%	WB	27.1	27.4	27.4	26.9	26.0	24.4	23.5	23.0	23.2	24.3	25.7	26.6
		MCDB	30.7	30.7	30.7	30.0	29.0	26.9	26.1	25.7	26.2	27.2	28.8	30.2
	10%	WB	26.7	27.1	27.0	26.5	25.5	24.0	23.1	22.6	22.8	23.9	25.2	26.2
		MCDB	30.1	30.2	30.1	29.4	28.3	26.5	25.6	25.3	25.7	26.7	28.2	29.4
Mean Daily Temperature Range	5% DB	MDBR	4.5	4.2	4.1	4.0	3.5	3.0	3.1	3.5	3.6	3.8	4.2	4.5
		MCDBR	5.7	5.2	5.1	4.9	4.6	4.1	4.0	4.3	4.5	4.6	5.2	5.7
		MCWBR	2.4	2.3	2.3	2.4	2.3	2.3	2.2	2.4	2.3	2.3	2.4	2.4
	5% WB	MCDBR	5.1	4.7	4.7	4.6	4.1	3.6	3.6	4.0	4.2	4.3	4.8	5.1
		MCWBR	2.4	2.3	2.3	2.5	2.5	2.2	2.2	2.4	2.4	2.4	2.4	2.4
Clear-Sky Solar Irradiance	taub	0.409	0.426	0.415	0.403	0.389	0.374	0.378	0.377	0.383	0.394	0.394	0.394	
	taud	2.538	2.484	2.517	2.542	2.562	2.559	2.544	2.539	2.506	2.495	2.525	2.554	
	Ebn at Noon	936	911	900	876	854	850	856	888	916	930	943	950	
	Edh at Noon	111	116	109	100	92	89	93	99	108	114	112	109	
All-Sky Solar Radiation	RadAvg	6.93	6.38	6.28	5.52	4.83	4.43	4.60	5.26	6.16	6.81	7.29	7.29	
	RadStd	0.53	0.54	0.26	0.34	0.25	0.16	0.10	0.15	0.15	0.26	0.22	0.31	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page